



SYDNEY BOYS HIGH  
MOORE PARK, SURRY HILLS

**2003**

**YEAR 11 YEARLY EXAMINATION**

# Mathematics

## General Instructions

- Reading time — 5 minutes
- Working time — 2 hours
- Write using black or blue pen
- Board-approved calculators may be used
- All necessary working should be shown in every question
- Start each Section in a new booklet

## Total marks — 130

- Attempt all questions
- Questions are not of equal value, the mark value is shown beside each part.
- Hand up your paper in five parts:
  - Section A, Questions 1 & 2;
  - Section B, Questions 3 & 4;
  - Section C, Questions 5 & 6;
  - Section D, Questions 7 & 8;
  - Section E, Questions 9 & 10

Examiner: R.Boros

**Total marks – 130**  
**Attempt Questions 1–10**

Answer each Section in a SEPARATE writing booklet. Extra writing booklets are available upon request.

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**Section A** Use a SEPARATE writing booklet.

**Marks**

**Question 1** (16 marks)

(a) Express each of the following as a rational number.

(i)  $49^{-\frac{1}{2}} \times 27^{\frac{2}{3}}$  1

(ii) The quotient of  $\sqrt{7}$  and  $\sqrt{63}$  1

(iii)  $\log_2 8$  1

(iv)  $\frac{\sqrt{32} - \sqrt{8}}{3\sqrt{2}}$  2

(b) Find, correct to 2 decimal places,  $\frac{(3 \cdot 24)^2}{5 \cdot 73 - 2 \cdot 84}$ . 1

(c) Solve for  $x$ ,  $\frac{2x}{x-5} = \frac{3}{5}$ . 2

(d) Factorise fully the expression  $x^3 - x^2 - x + 1$ . 2

(e) Find the centre and radius of the circle  $x^2 + y^2 - 6x + 4y - 12 = 0$ . 2

(f) Solve the inequality  $x^2 - 4x < 0$ . 2

(g) Given  $v^2 = u^2 - 2aS$ ,  $v = 2 \cdot 5$ ,  $u = 2 \cdot 3$ , and  $a = 7$ , find  $S$  correct to 3 significant figures. 2

**Section A continued**

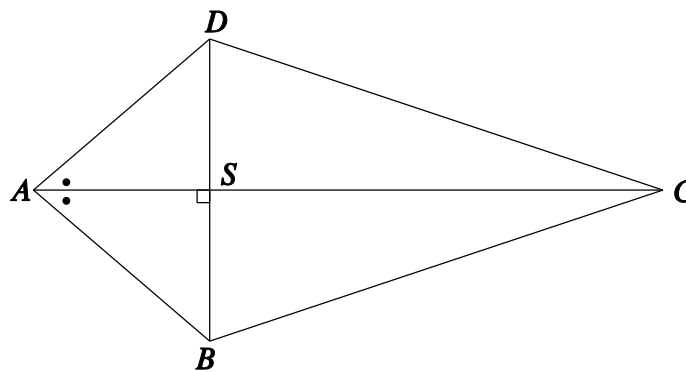
**Marks**

**Question 2** (14 marks)

(a) Given the points  $A(-5, 3)$ ,  $B(1, -5)$ , and  $C(2, 2)$ ,

- |                                                                     |          |
|---------------------------------------------------------------------|----------|
| (i) Find the length of $AB$ .                                       | <b>1</b> |
| (ii) Find the equation of the line $AB$ written in general form.    | <b>2</b> |
| (iii) Find the perpendicular distance of $C$ from the line $AB$ .   | <b>2</b> |
| (iv) Hence or otherwise, calculate the area of the triangle $ABC$ . | <b>1</b> |

(b)



In the above diagram, not to scale,  $ABCD$  is a quadrilateral. The diagonals  $AC$  and  $DB$  intersect at right angles at point  $S$ .  $\hat{DAS} = \hat{BAS}$ .

- |                                                                  |          |
|------------------------------------------------------------------|----------|
| (i) Prove that $\triangle ASB$ is congruent to $\triangle ASD$ . | <b>2</b> |
| (ii) Hence prove that $DA = BA$ .                                | <b>1</b> |
- (c) Given  $f(x) = x^2 + 3x + 2$ ,
- |                                              |          |
|----------------------------------------------|----------|
| (i) Evaluate $f(-3)$ .                       | <b>1</b> |
| (ii) Find a simple expression for $f(a+2)$ . | <b>2</b> |
- (d) Solve for  $x$ ,  $|2x - 1| \leq 5$ .
- 2**

**Section B** Use a SEPARATE writing booklet.

**Marks**

**Question 3** (13 marks)

- (a) Doug observes a clifftop  $A$  from his yacht at position  $P$ . He then sails 500 m towards the cliff to position  $Q$ . The angle of elevation to the clifftop from  $P$  is  $5^\circ$  and from  $Q$  is  $8^\circ$ .
- (i) Draw a diagram to illustrate the above information and use the Sine Rule to calculate  $AQ$  correct to the nearest metre. 2
- (ii) Hence or otherwise find the distance  $QB$  correct to the nearest 10 metres. 2
- (b) Simplify the expression  $\frac{\tan\theta}{\cot\theta} - \frac{\sec^2\theta}{1}$ . 2
- (c) (i) Find  $\theta$  given that  $\sin\theta = \frac{1}{2}$  where  $0^\circ \leq \theta \leq 180^\circ$ . 1
- (ii) Hence find the exact values of  $\tan \theta$  and  $\sec \theta$ . 2
- (d) Simplify  $\sin\theta \cos(90^\circ - \theta) + \cos\theta \sin(90^\circ - \theta)$ . 2
- (e) If  $\sin\theta = \frac{8}{17}$  and  $\theta$  is an acute angle, find the exact values of  $\cos \theta$  and  $\tan \theta$ . 2

**Section B continued****Marks****Question 4** (15 marks)

- (a) Sketch the area defined by the inequality  $y \leq x^2$ . **2**
- (b) Let  $\alpha$  and  $\beta$  be the roots of the equation  $x^2 - 5x + 2 = 0$ . Find the values of:
- (i)  $\alpha + \beta$  **1**
- (ii)  $\alpha \beta$  **1**
- (iii)  $(\alpha + 1)(\beta + 1)$  **1**
- (c) State the domain and range of each of these functions.
- (i)  $x^2 + 3 = y$  **2**
- (ii)  $y = 3^x$  **2**
- (d) Solve for  $x$  the equation  $9^x - 9(3)^x = 0$ . **3**
- (e) The roots of the quadratic equation  $px^2 - x + q = 0$  are  $-1$  and  $3$ . Find  $p$  and  $q$ . **3**

**Section C** Use a SEPARATE writing booklet.

**Marks**

**Question 5** (15 marks)

- (a) A parabola has the equation  $y = x^2 - 12x + 20$ . Find
- (i) where it cuts the  $x$  and  $y$  axes, 2
  - (ii) its axis of symmetry and vertex, 2
  - (iii) the focus, by first expressing it in the form  $(x - h)^2 = 4a(y - k)$ , 2
  - (iv) the equation of the directrix. 1
- (b) For each of the quadratics below, evaluate the discriminant and state the relevance of this with regard to the roots of the equation.
- (i)  $x^2 + 3x + \frac{9}{4} = y$  2
  - (ii)  $3x^2 - 2x = y + 5$  2
- (c) Find the values of  $M$  for which the equation  $4x^2 - Mx + 9 = 0$  has
- (i) exactly one real root, 2
  - (ii) real roots. 2

**Section C continued****Marks****Question 6** (14 marks)

- (a) Solve the following equations simultaneously,  
 $4x - y = 3$  and  $10x + 3y = 2$ . **2**
- (b) Find  $x$  given  $2\log_9\sqrt{3} + \log_9 81 = x$ . **3**
- (c) Find  $x$  correct to 3 decimal places given that  $7^x = 15$ . **2**
- (d) If Ron invests \$500 at 12.5 % p.a. compound interest, how long would it take the investment to grow to a sum of \$1000. (Answer in years, correct to 2 decimal places.) **3**
- (e) The  $r^{\text{th}}$  term of a series is  $3 \times 2^{(r-4)}$ . Determine which of the numbers 96,  $\frac{3}{4}$ , 256 belong to the series. **2**
- (f) Evaluate  $\sum_{n=-1}^7 (2n + 3)$ . **2**

**Section D** Use a SEPARATE writing booklet.

**Marks**

**Question 7** (10 marks)

- (a) Let  $A(0, -2)$  and  $B(1, 0)$  be 2 fixed points and let  $P(x, y)$  be a variable point. **3**  
Find the locus of  $P$  such that the length  $(PA)^2$  equals the length  $(PB)^2$ .
- (b) Find the locus of points 2 units away from the line  $y = 3$ . **2**
- (c) Sketch the graph of  $y = \sin \theta$  given  $-180^\circ < \theta < 90^\circ$ . **3**
- (d) Find the values of  $a$  and  $b$  given that **2**  
$$(ax - 3)^2 + b = 4x^2 - 12x + 15.$$

**Section D continued****Marks****Question 8** (12 marks)

(a) Differentiate each of the following:

(i)  $y = 2x^3 - 8$  **1**

(ii)  $y = (2x - 1)^3$  **1**

(iii)  $y = \frac{2x}{1-3x}$  **2**

(iv)  $y = x^2\sqrt{x}$  **2**

(v)  $y = \frac{7}{2x^3}$  **1**

(b) (i) Find the gradient of the normal to the curve  $y = 1 - \frac{1}{2}x^2$  at the point (1, 3). **3**

(ii) Find the point on  $y = 1 - \frac{1}{2}x^2$  where the tangent to the curve is parallel to this normal. **2**

**Section E** Use a SEPARATE writing booklet.

**Marks**

**Question 9** (11 marks)

(a) Evaluate the following limits:

(i)  $\lim_{x \rightarrow 5} \left( \frac{x-5}{x^2-25} \right)$  **2**

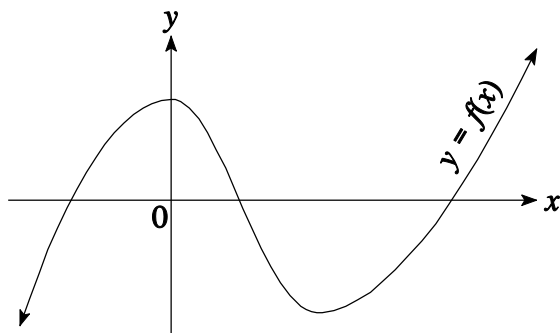
(ii)  $\lim_{x \rightarrow \infty} \left( \frac{7-2x-3x^2}{5x^2+3} \right)$  **3**

(b) Two cars depart town *A* at the same time. Car *X* travels at 60 km/h on a bearing of  $345^\circ\text{T}$ , whilst car *Y* travels at 100 km/h on a bearing of  $085^\circ\text{T}$ . How far apart would the cars be after 3 hours? Answer correct to one decimal place. **3**

(c) Differentiate  $y = x^2 + x$  from first principles. **3**

**Section E continued****Marks****Question 10** (10 marks)

- (a) The diagram below shows  $y = f(x)$ .

**3**

Copy or trace this diagram into your answer booklet and sketch a possible graph for  $y = f'(x)$  on the same set of axes.

- (b) A farmer wishes to make a rectangular enclosure using a river as one boundary and 400 m of fencing on the other three sides.

- (i) Find the maximum possible area of the enclosure.

**3**

- (ii) What are the dimensions of this enclosure?

**1**

- (c) Find  $x$  correct to 3 decimal places, given that  $\log_7 6 - 2\log_7 3 = x$ .

**3****END OF THE PAPER**



**SYDNEY BOYS HIGH SCHOOL**  
MOORE PARK, SURRY HILLS

**SEPTEMBER 2003**

**YEARLY EXAMINATION**

**YEAR 11**

**Mathematics**

**Sample Solutions**

Question (1) [16 marks]

(a)  $49^{-1/2} \times 27^{2/3}$

(i)  $= \frac{1}{\sqrt{49}} \times (3^3)^{2/3}$  ①

(ii)  $\frac{\sqrt{7}}{\sqrt{63}} = \frac{1}{3}$  ①

(iii)  $\log_2 8 = 3 \log_2 2 = 3$  ①

(iv)  $\frac{2\sqrt{2}(2-1)}{3\sqrt{2}} = \frac{2}{3}$  ①

(b)  $\frac{(3 \cdot 24)^2}{5 \cdot 73 - 2 \cdot 84} = \frac{3 \cdot 63}{(2d.f.)}$

(c)  $\frac{2x}{x-5} = \frac{3}{5}$

$10x = 3x - 15$

$7x = -15$

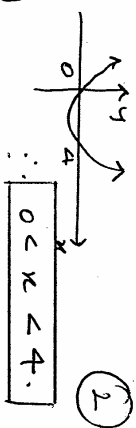
(d)  $\frac{x^2(x-1) - (x-1)}{(x-1)^2(x+1)}$  ②

(e)  $x^2 - 6x + 9 + y^2 + 4y + 4 = 25$  ②

$\therefore (x-3)^2 + (y+2)^2 = 25$

Centre (3, -2) r = 5

(f)  $x^2 - 4x < 0$   
 $x(x-4) < 0$



(g)  $(2.5)^2 = (2.3)^2 - 14(s)$  ②  
 $14s = 0.96$   
 $s = -0.0686$  (35.f.)

Question (2) [14 marks]

(a) (i)  $AB = \sqrt{(-6)^2 + 8^2} = 10$  ①

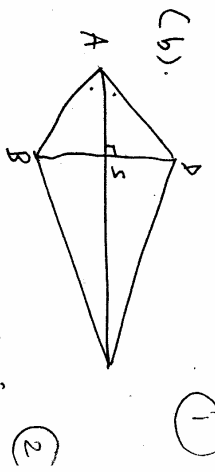
(ii)  $w = -\frac{8}{6} = -\frac{4}{3}$

$y - 3 = -\frac{4}{3}(x + 5)$  ②

$3y - 9 = -4x - 20$   
 $4x + 3y + 11 = 0$

(iii)  $d = \frac{1}{\sqrt{25}} \sqrt{8^2 + 11^2} = \frac{\sqrt{173}}{5}$  ②

(iv)  $A = \frac{1}{2} \times 10 \times \frac{2\sqrt{5}}{5} = 2\sqrt{5}$  ①



$\angle DAS = \angle CBS$  (given) ②  
 $\angle DSA = \angle BSA$  (st. line)  
AS is common

$\therefore \triangle DAS \cong \triangle CBS$  (AAS)

$\therefore DA = CB$   
(conv. sides of cong.  $\Delta$ s) ①

(c)  $f(x) = x^2 + 3x + 2$

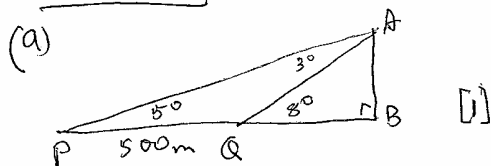
(i)  $f(-3) = 2$  ①

(ii)  $(a+2)^2 + 3(a+2) + 2 = a^2 + 7a + 12$  ②

(d)  $(a+3)(a+d) = a^2 + 7a + 12$  ②

$2x - 1 \leq 5$   
 $-5 \leq 2x - 1 \leq 5$   
 $-4 \leq 2x \leq 6$   
 $-2 \leq x \leq 3$  ②

### Question 3



(i)  $\angle PAQ = 3^\circ$  (Ext. angle thm)

$$\frac{AQ}{\sin 5^\circ} = \frac{500}{\sin 3^\circ}$$

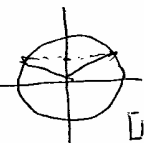
$$AQ = \frac{500 \sin 5^\circ}{\sin 3^\circ} \approx 833 \text{ m} \quad [1]$$

(ii)  $\cos 8^\circ = \frac{QB}{AQ}$

$$\therefore QB = AQ \cos 8^\circ \approx 820 \text{ m (nearest 10m)} \quad [2]$$

(b)  $\frac{\tan \theta}{\cot \theta} - \sec^2 \theta$   
 $= \tan^2 \theta - \sec^2 \theta$   
 $= \tan^2 \theta - (1 + \tan^2 \theta)$   
 $= -1 \quad [2]$

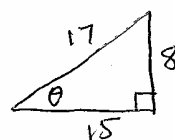
(c) (i)  $\sin \theta = \frac{1}{2} \quad 0^\circ \leq \theta \leq 180^\circ$

$\theta = 30^\circ \text{ or } 150^\circ$  

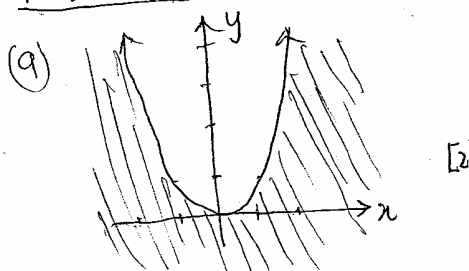
(ii)  $\tan \theta = \frac{1}{\sqrt{3}} \text{ or } -\frac{1}{\sqrt{3}}$   
 $\sec \theta = \frac{2}{\sqrt{3}} \text{ or } -\frac{2}{\sqrt{3}} \quad [2]$

(d)  $\sin \theta \cdot \cos(90^\circ - \theta) + \cos \theta \sin(90^\circ - \theta)$   
 $= \sin \theta \cdot \sin \theta + \cos \theta \cdot \cos \theta$   
 $= \sin^2 \theta + \cos^2 \theta \quad [2]$   
 $= 1$

(e)  $\sin \theta = \frac{8}{17}$   
 $\cos \theta = \frac{15}{17} \quad [2]$   
 $\tan \theta = \frac{8}{15}$



### Question 4



(b)  $x^2 - 5x + 2 = 0$

(i)  $\alpha + \beta = 5 \quad [1]$

(ii)  $\alpha \beta = 2 \quad [1]$

(iii)  $(\alpha + 1)(\beta + 1) = \alpha\beta + (\alpha + \beta) + 1$   
 $= 2 + 5 + 1$   
 $= 8 \quad [1]$

(c) (i)  $y = x^2 + 3 \quad D = \{x: x \in \mathbb{R}\}$   
 $R = \{y: y \geq 3\}$

(ii)  $y = 3^x \quad D = \{x: x \in \mathbb{R}\}$   
 $R = \{y: y > 0\}$

Question 4 (cont'd)

(d)  $9^x - 9(3)^x = 0$

$(3^x)^2 - 9(3^x) = 0$

Let  $u = 3^x$

$u^2 - 9u = 0$

$u(u - 9) = 0$

$\therefore u = 0$  or  $u = 9$

$\therefore 3^x = 0$  or  $3^x = 9$

No soln  $\underline{\underline{x = 2}}$  [3]

(e)  $px^2 - x + q = 0$

Roots -1 and 3

$\alpha + \beta = \frac{1}{p}$      $\alpha \beta = \frac{q}{p}$

|                              |                    |
|------------------------------|--------------------|
| $\therefore \frac{1}{p} = 2$ | $\frac{q}{p} = -3$ |
| $p = \frac{1}{2}$            | $2q = -3$          |
| <hr/>                        | $q = -\frac{3}{2}$ |

[3]

- [2] 5. (a) (i) On the  $x$ -axis,  $y = 0$ ; so  $(x - 10)(x - 2) = 0$ .  
On the  $y$ -axis,  $x = 0$ ; so  $y = 20$ .  
 $\therefore$  The curve meets the  $x$ -axis at 2, 10, and the  $y$ -axis at 20.
- [2] (ii) The midpoint of 2 and 10 is 6, so the axis of symmetry is  $x = 6$ .  
When  $x = 6$ ,  $y = -16$ , so the vertex is  $(6, -16)$ .
- [2] (iii)  $x^2 - 12x + 36 = y - 20 + 36$ ,  
 $(x - 6)^2 = 4 \times \frac{1}{4} \times (y + 16)$ .  
 $\therefore$  The focus is at  $(6, -15\frac{3}{4})$ .
- [1] (iv) The directrix is  $y = -16\frac{1}{4}$ .
- [2] (b) (i)  $\Delta = 9 - 9 = 0 \implies$  one real root.
- [2] (ii)  $\Delta = 4 + 60 = 64 \implies$  two real, rational roots.
- [2] (c) (i)  $\Delta = M^2 - 144 = 0$ ,  
 $M^2 = 144$ .  
 $\therefore$  One root when  $M = \pm 12$ .
- [2] (ii)  $M^2 - 144 \geq 0$ ,  
 $M^2 \geq 144$ ,  
 $|M| \geq 12$ .  
 $\therefore M \leq -12, M \geq 12$ .
- [2] 6. (a)  $4x - y = 3, \dots\dots\dots(1)$   
 $10x + 3y = 2. \dots\dots\dots(2)$   
 $y = 4x - 3$  from (1), then substitute in (2).  
 $10x + 3(4x - 3) = 2$ ,  
 $10x + 12x - 9 = 2$ ,  
 $22x = 11$ ,  
 $x = \frac{1}{2}$ ,  
 $y = -1$ .
- [3] (b)  $\log_9 3 + \log_9 9^2 = x$ ,  
 $\log_9 9^{\frac{1}{2}} + \log_9 9^2 = x$ ,  
 $\frac{1}{2} \log_9 9 + 2 \log_9 9 = x$ ,  
 $\frac{1}{2} + 2 = x$ ,  
*i.e.*,  $x = 2\frac{1}{2}$  (or  $\frac{5}{2}$ ).
- [2] (c)  $\log 7^x = \log 15$ ,  
 $x \log 7 = \log 15$ ,  
 $x = \frac{\log 15}{\log 7}$ ,  
 $\approx 1.392$ .

$$\begin{aligned}
\boxed{3} \quad (d) \quad & \$1\,000 = \$500 \left(1 + \frac{12.5}{100}\right)^n, \\
& 2 = (1.125)^n, \\
& \log 1.125^n = \log 2, \\
& n \log 1.125 = \log 2, \\
& n = \frac{\log 2}{\log 1.125} \\
& n \approx 5.88 \text{ years.}
\end{aligned}$$

$$\begin{aligned}
\boxed{2} \quad (e) \quad & 96 \div 3 = 32, \quad \frac{3}{4} \div 3 = \frac{1}{4}, \quad 256 \div 3 = 85\frac{1}{3}, \\
& = 2^5, \quad = 2^{-2}, \quad \text{Not a power of 2.} \\
& \therefore 96, \frac{3}{4} \text{ belong to the series.}
\end{aligned}$$

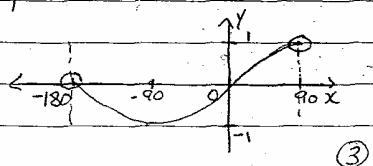
$$\begin{aligned}
\boxed{2} \quad (f) \quad & \sum_{n=-1}^7 (2n+3) = 1 + 3 + 5 + 7 + 9 + 11 + 13 + 15 + 17, \\
& = 81.
\end{aligned}$$

# QUESTION 7

(a)  $A(0, -2) B(1, 0) (PA)^2 = (PB)^2$   
 $(x-0)^2 + (y+2)^2 = (x+1)^2 + (y-0)^2$   
 $x^2 + y^2 + 4y + 4 = x^2 + 2x + 1 + y^2$   
 $4y + 2x + 3 = 0$   
 $2x + 4y + 3 = 0$  (3)

(b)  $y = 5$  or  $y = 1$  (2)

(c)  $y = \sin \theta \quad -180^\circ < \theta < 90^\circ$



(d)  $(ax-3)^2 + b = 4x^2 - 12x + 15$   
 $a^2x^2 - 6ax + 9 + b = 4x^2 - 12x + 15$   
 equating coefficients  
 $a^2 = 4 \quad -6ax = -12x \quad 9+b=15$   
 $a = \pm 2 \quad a = 2 \quad b = 6$   
 $a = 2, b = 6$   
 or  
 $(ax-3)^2 + b = 4x^2 - 12x + 9 + 6$   
 $(ax-3)^2 + b = (2x-3)^2 + 6$   
 $a = 2 \quad b = 6$  (2)

# QUESTION 8

(a)(i)  $y = 2x^3 - 8$   
 $y' = 6x^2$  (1)

(ii)  $y = (2x-3)^3$   
 $y' = 3(2x-3)^2 \times 2$   
 $= 6(2x-3)^2$  (1)

(iii)  $y = \frac{2x}{1-3x}$   
 $y' = \frac{vdu - u dv}{v^2}$

$= \frac{(1-3x) \times 2 - 2x \times -3}{(1-3x)^2}$

$= \frac{2}{(1-3x)^2}$  (2)

(iv)  $y = x^2 \sqrt{x}$   
 $= x^{\frac{5}{2}}$   
 $y' = \frac{5}{2} x^{\frac{3}{2}}$   
 $= \frac{5}{2} \sqrt{x^3}$  (2)

(v)  $y = \frac{7}{2x^3} = \frac{7}{2} x^{-3}$   
 $y' = -\frac{21}{2} x^{-4}$  (1)  
 $= -\frac{21}{2x^4}$

(b)  $y = 1 - \frac{1}{2} x^2$   
 (i)  $y' = -x$   
 when  $x = 1$  grad of tangent = -1  
 " normal = 1 (3)

(ii)  $y' = -x$   
 $-x = 1$  when  $x = -1$   
 point  $(-1, \frac{1}{2})$  (2)

**Question 9.**

$$\begin{aligned}
 (a) \quad (i) \quad & \lim_{x \rightarrow 5} \frac{x-5}{x^2-25} \\
 &= \lim_{x \rightarrow 5} \frac{x-5}{(x-5)(x+5)} \\
 &= \lim_{x \rightarrow 5} \frac{1}{(x+5)} \\
 &= \frac{1}{10}
 \end{aligned}$$

$$\begin{aligned}
 (ii) \quad & \lim_{x \rightarrow \infty} \frac{7-2x-3x^2}{5x^2+3} \\
 &= \lim_{x \rightarrow \infty} \frac{\frac{7}{x^2} - \frac{2}{x} - \frac{3x^2}{x^2}}{\frac{5x^2}{x^2} + \frac{3}{x^2}} \\
 &= -\frac{3}{5}
 \end{aligned}$$

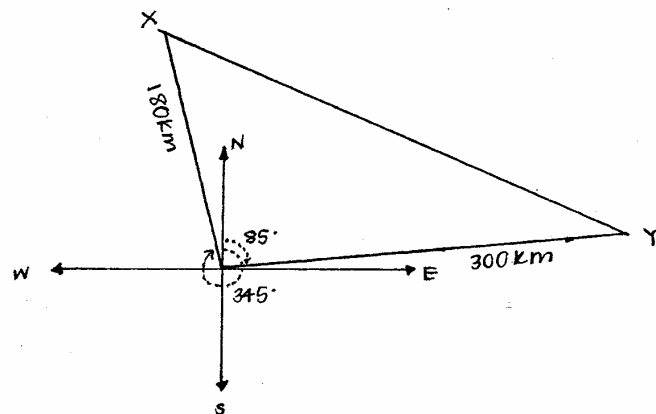
- (b) In 3 hours car X travels 180km.  
In 3 hours car Y travels 300km.

$$\begin{aligned}
 XY^2 &= AY^2 + AX^2 - 2AY \cdot AX \cdot \cos A \\
 &= 300^2 + 180^2 - 2(300)(180)\cos 100^\circ
 \end{aligned}$$

$$\begin{aligned}
 XY^2 &= AY^2 + AX^2 - 2AY \cdot AX \cdot \cos A \\
 &= 300^2 + 180^2 - 2(300)(180)\cos 100^\circ
 \end{aligned}$$

$$XY = 375.7$$

$$\therefore XY = 375.7 \text{ km (1d.p.)}$$



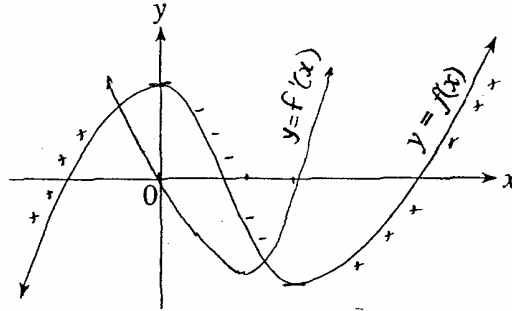
(c) Let  $f(x) = x^2 + x$   
 Let  $f(x+h) = (x+h)^2 + (x+h)$

$$\begin{aligned}
 f'(x) &= \lim_{h \rightarrow \infty} \frac{f(x+h) - f(x)}{h} \\
 &= \lim_{h \rightarrow \infty} \frac{(x+h)^2 + (x+h) - (x^2 + x)}{h} \\
 &= \lim_{h \rightarrow \infty} \frac{x^2 + 2xh + h^2 + x + h - x^2 - x}{h} \\
 &= \lim_{h \rightarrow \infty} \frac{2xh + h^2 + h}{h} \\
 &= \lim_{h \rightarrow \infty} \frac{h(2x + h + 1)}{h} \\
 &= 2x + 1
 \end{aligned}$$

$\therefore f'(x) = 2x + 1$  by first principles.

**Question 10.**

(a)



(b) (i) Let  $P = 2x + y$ , where  $P = 400$ .

$$400 = 2x + y$$

$$\therefore y = 400 - 2x$$

$$\text{Area} = xy$$

$$= x(400 - 2x)$$

$$= 400x - 2x^2$$

Using the axis of symmetry formula,  $x = -\frac{b}{2a}$ , where  $a = -2, b = 400$ .

$$\begin{aligned} x &= -\frac{b}{2a} \\ &= -\frac{400}{2(-2)} \\ &= 100 \end{aligned}$$

At  $x = 100, y = 200$ .

Hence the maximum possible area of this enclosure is  $A = 20000 \text{ m}^2$

(ii) Hence the dimensions of this enclosure are 100m by 200m.

(c)

$$\log_7 6 - 2\log_7 3 = x$$

$$\log_7 6 - \log_7 3^2 = x$$

$$\frac{\log_{10} 6}{\log_{10} 7} - \frac{\log_{10} 9}{\log_{10} 7} = x$$

$$0.9207 - 1.1291 = x$$

$$x = -0.20845$$

$$\therefore x = -0.208 \text{ (3dp).}$$

or

$$\log_7 \left( \frac{6}{3^2} \right) = x$$

$$\log_7 \left( \frac{2}{3} \right) = x$$

$$7^x = \frac{2}{3}$$

$$x \log 7 = \log \frac{2}{3}$$

$$\therefore x = -0.208$$